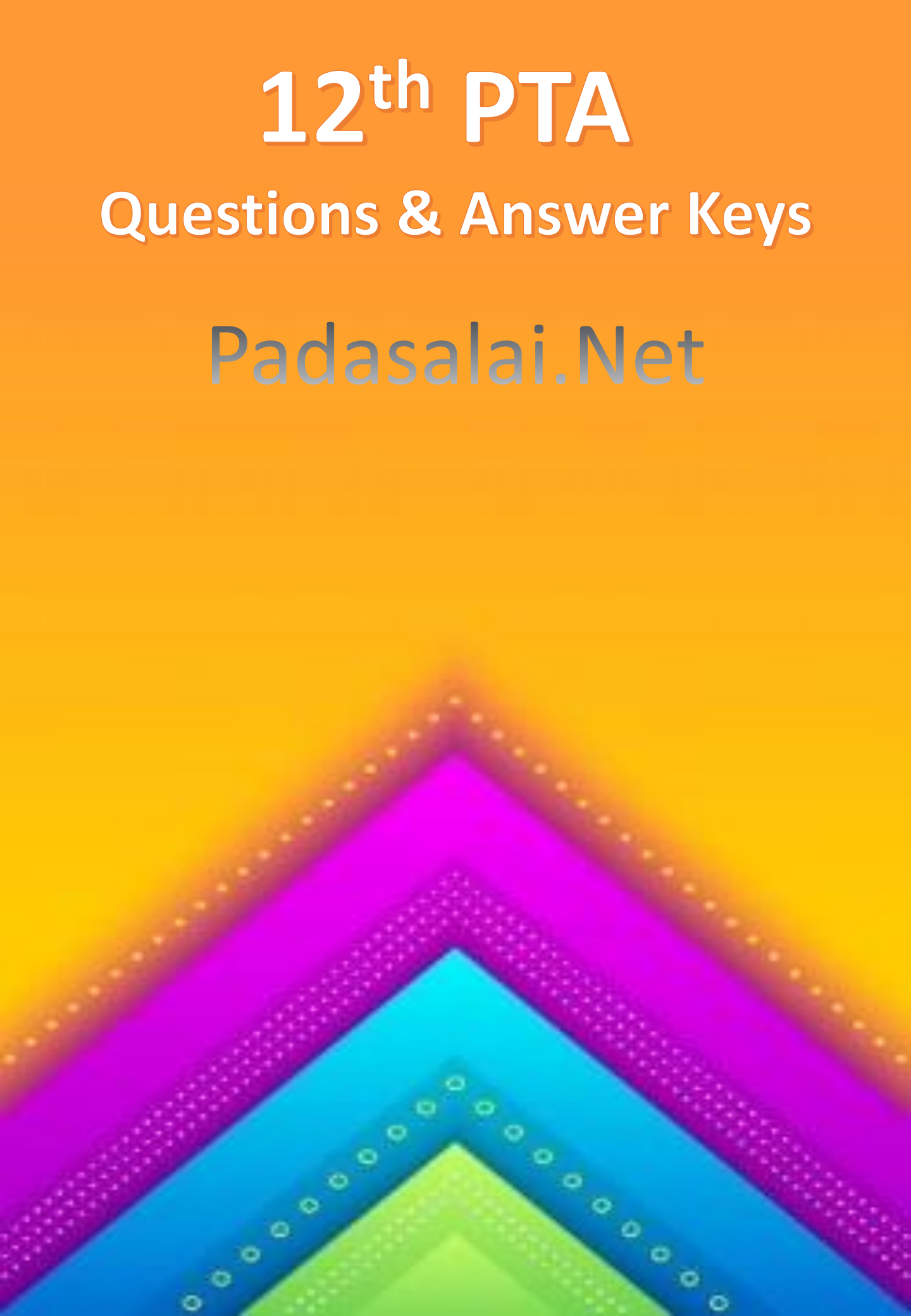
12th PTA

Questions & Answer Keys

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PTA – PHYSICS - MODEL QUESTION PAPER - 6
STANDARD XII

TIME ALLOWED : 15min + 3.00 Hr **Maximum Marks : 70**

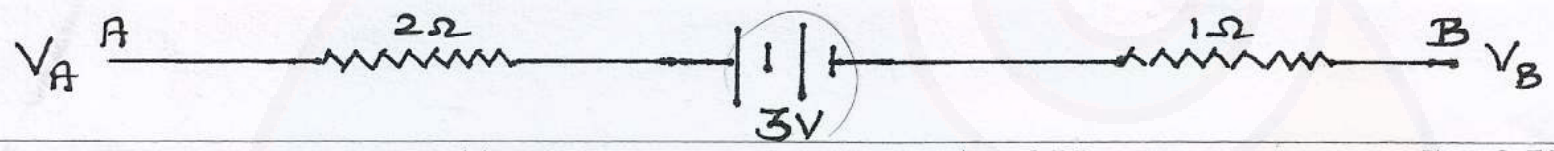
Instructions: 1. Check the questions paper for fairness of printing. If there is any lack of fairness, inform the hall supervisor immediately.
2. Use Blue or Black ink to write and underline use pencil to draw diagrams

PART – I

Note: (i) Answer all the questions **15× 1 =15**
(ii) Choose the most appropriate answer from four given alternatives and write the option code with the corresponding answer

1. An electric dipole is placed at an angle 30° with an electric field intensity of 2×10^5 NC^{-1}. It experiences a torque equal to 4 Nm. The charge on the dipole if the dipole length is 2 cm, is
a) 8 mC **b) 2 mC** c) 5mC d) 7 mC
Hints: $\tau = p\,E\sin\theta = q\times 2a\times E\times\sin\theta$ $\therefore q = \frac{\tau}{2a\times E\times\sin\theta} = \frac{4}{2\times10^{-2}\times2\times10^{5}\times\sin 30^\circ} = 2mC$

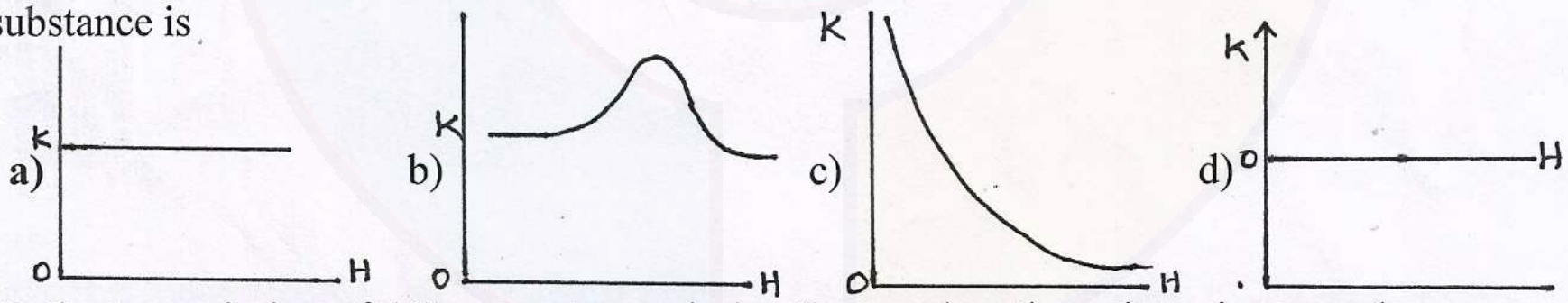
2. The potential difference between the points A and B in the given figure is



a) – 3 V b) +3 V c) +6 V **d) +9 V**
Hints: $V_A - V_B = \xi + IR_1 + IR_2 = 3 + (2\times 2) + (1\times 2) = 3+4+2 = 9$ V

3. A vertical straight conductor carries a current vertically upwards . A point P lies to the east of it at a small distance and another point Q lies to the west at the same distance. The magnetic field at P is
a) greater than that at Q **b) same as at Q** c) less than at Q
d) greater or less than at Q depending upon strength of current

4. The variation of magnetic susceptibility (k) with magnetising field (H) for a paramagnetic substance is



5. In the transmission of AC power transmission lines , when the voltage is stepped up n times , the power loss in transmission
a) Increases n times b) decreases n times c) Increases n^2 times **d) decreases n^2 times**
Hints: $P = I^2R = \frac{V^2}{R}$, If V increases to nV power loss = $P = \frac{n^2V^2}{R}$

6. Maxwell's modified form of Ampere's circuital law is
a) $\oint \vec{B}.\vec{ds} = 0$ b) $\oint \vec{B}.\vec{dl} = \mu_o I$ c) $\oint \vec{B}.\vec{dl} = \mu_o I + \frac{1}{\varepsilon_o}\frac{dq}{dt}$ **d) $\oint \vec{B}.\vec{dl} = \mu_o I + \mu_o \varepsilon_o \frac{d\varphi}{dt}$**

7. The transverse nature of light is shown in
a) interference b) diffraction c) scattering **d) polarisation**

8. The magnification of a telescope is given by
a) $\frac{f_e}{f_o}$ **b) $\frac{f_o}{f_e}$** c) $2\frac{f_e}{f_o}$ d) $\frac{f_o+f_e}{2}$

9. The threshold wavelength for a metal surface whose photoelectric work function is 2.063 eV
a) 4125Å b) 3750 Å **c) 6000 Å** d) 2062.5
Hints: $\phi = \frac{hC}{\lambda_o}$; $\lambda_o = \frac{hC}{\phi} = \frac{6.626\times10^{-34}\times3\times10^{8}}{2.063\times1.6\times10^{-19}} = 6.022\times10^{-7} \approx 6000$ Å

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10. Which of the following transition will have highest emission wavelength

a) n = 2 to n = 1 b) n = 4 to n = 1 c) n = 6 to n = 2 **d) n = 5 to n = 2**

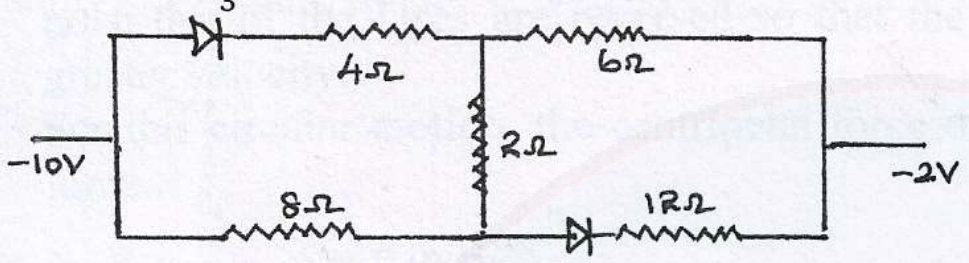
11. A system of N_o nucleus at t=0, then number of nuclei remaining after half of a half life

$(t = \frac{1}{2}T_{\frac{1}{2}})$ a) $\frac{N_0}{2}$ **b)** $\frac{N_0}{\sqrt{2}}$ c)) $\frac{N_0}{4}$ d) $\frac{N_0}{8}$

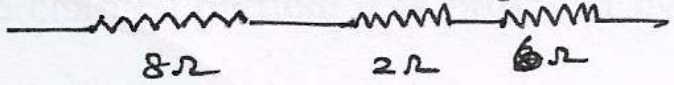
Hints: $\frac{N}{N_0} = \left(\frac{1}{2}\right)^n$, the number of half lives.= $n = \frac{\text{Total life}}{\text{Half life}} = \frac{\frac{1}{2}T_{\frac{1}{2}}}{T_{\frac{1}{2}}} = \frac{1}{2}$; $\therefore \frac{N}{N_0} = \left(\frac{1}{2}\right)^{\frac{1}{2}} = \frac{1}{\sqrt{2}}$; $N = \frac{N_0}{\sqrt{2}}$

12. In the above circuit the equivalent resistance between A and B is

a) $\frac{20}{3}\ \Omega$ b) 10 Ω **c) 16 Ω** d) 20 Ω



Hints:- Diodes are in Reverse bias current flows through



= 16Ω

13. If the input to the NOT gate is A= 1011, then its output is

a) 0100 b)1000 **c) 1100** d) 0011

14. In the space wave propagation , the range of coverage of the propagation depends on the height(h) of the antenna given by the equation

a) $\sqrt{2Rh}$ b) $\sqrt{\frac{Rh}{2}}$ c) $\sqrt{2R^2h}$ d) $\frac{h}{2R}$

15. The strongest source of gravitational wave is

a) Sun b) Stars c) Neutron Star **d) black holes**

PART – II

Answer any six questions . Question number 24 compulsory. $6 \times 2 = 12$

16. Define electric potential. Give a relation

The electric potential at a point P is equal to the work done by an external force to bring a unit positive charge with constant velocity from infinity to the point P in the region of the external electric field $\vec{E}$. **unit** : volt

17. The resistance of nichrome wire at 0°C is 10 Ω. If its temperature coefficient of resistance is 0.004 / °C, find its resistance at boiling point of water. Comment on the result.

$R_t = R_0[1 + \alpha(T - T_0)]$

$= 10[1 + 0.004(100 - 0)] = 10(1 + 0.4) = 10 \times 1.4 = 14\ \Omega$

Temperature increases resistance also increases

18. State Ampere's circuital law. Give its mathematical form.

The line integral of magnetic field over a closed loop is μ_0 times net current enclosed by the loop. $\oint \vec{B} \cdot \overrightarrow{d\ell} = \mu_o I_{enclosed}$

where I enclosed is the net current linked by the closed loop C.

19. What is meant by mutual induction?

When an electric current passing through a coil changes with time, an emf is induced in the neighbouring coil. This phenomenon is known as mutual induction

20. What is displacement current?

It is defined as the current which comes into play in the region in which the electrifield is changing with time.

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21. Differentiate between real and virtual images.

real images	virtual images
1.They are inverted	They are erect
2.They are formed due to the actual intersection of light rays	They are formed due to imaginary intersection of light rays
3. They are formed on the screen	They are appeared to be on the lens or the mirror itself

22. Calculate the cut-off wavelength and cutoff frequency of x-rays from an x –ray tube of accelerating potential 20,000 V.

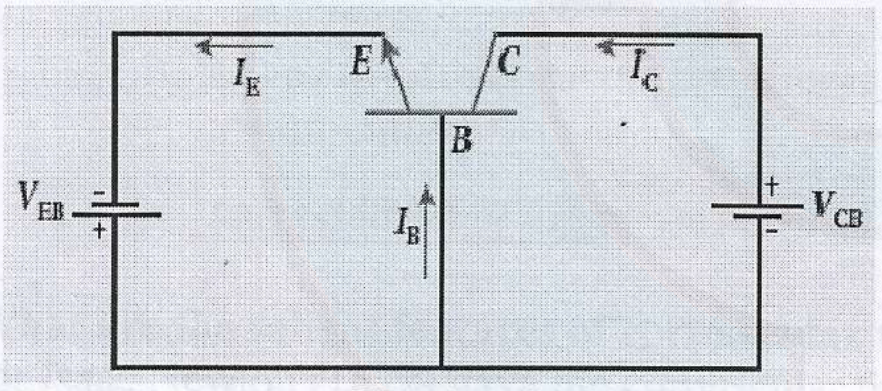
cut-off wavelength $\lambda_o = \frac{12400\text{Å}}{V} = \frac{12400\text{Å}}{20000} = 0.62\text{Å}$

$C = \nu_o \lambda_o$; cutoff frequency $\nu_o = \frac{c}{\lambda_o} = \frac{3\times10^8}{0.62\times10^{-10}} = 4.839$ Hz

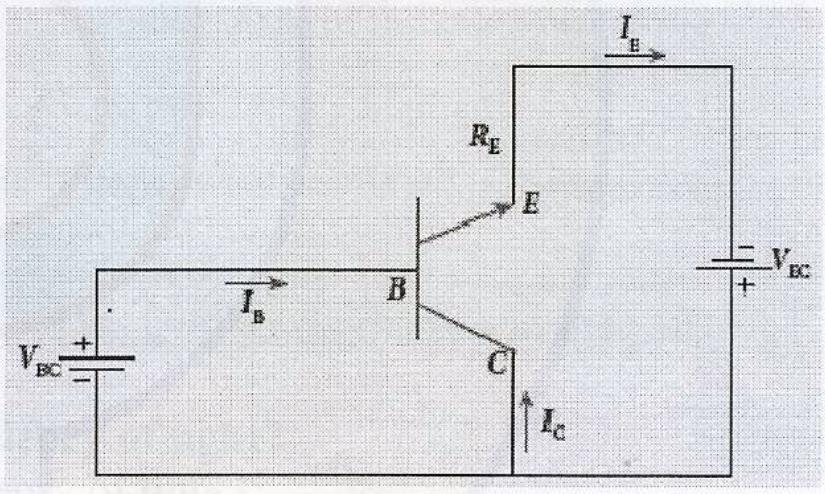
23. In a nuclear fission 0.1% mass is converted into energy. Calculate the energy released by the fission of 1kg mass.

$$E = mC^2 = \left(\frac{0.1}{100} \times 1\right) \times (3 \times 10^8)^2 = 10^{-3} \times 9 \times 10^{16} = 9 \times 10^{13}\ \text{J}$$

24. Draw the circuit diagrams of transistor in CB and CC modes.



CB circuit



CE circuit

PART - III

Answer any six questions . Question number 33 compulsory. $6 \times 3 = 18$

25. When two objects are rubbed with each other , approximate charge of 50nC can be produced in each subject. Calculate the number of electrons that must be transferred to produce this charge.

$q = ne$; $n = \frac{q}{e} = \frac{50\times10^{-9}}{1.6\times10^{-19}} = 3.125 \times 10^{11}$ electrons

26. Differentiate between Joule's heating effect and Peltier effect.

Joule's effect	Peltier effect
1. $H \propto I^2$	1. $H \propto I$
2. It is independent of direction of current When the direction of current changes , there is no change in the heating effect	2. It is dependent of direction of current If the direction of current changes , Peltier effect is reversed
3. It takes place throughout the conductor	3. It takes place only at the junction
4. Heat is evolved	4. Heat is evolved at one junction and absorbed at another junction
5. It is an irreversible process	5. It is a reversible process

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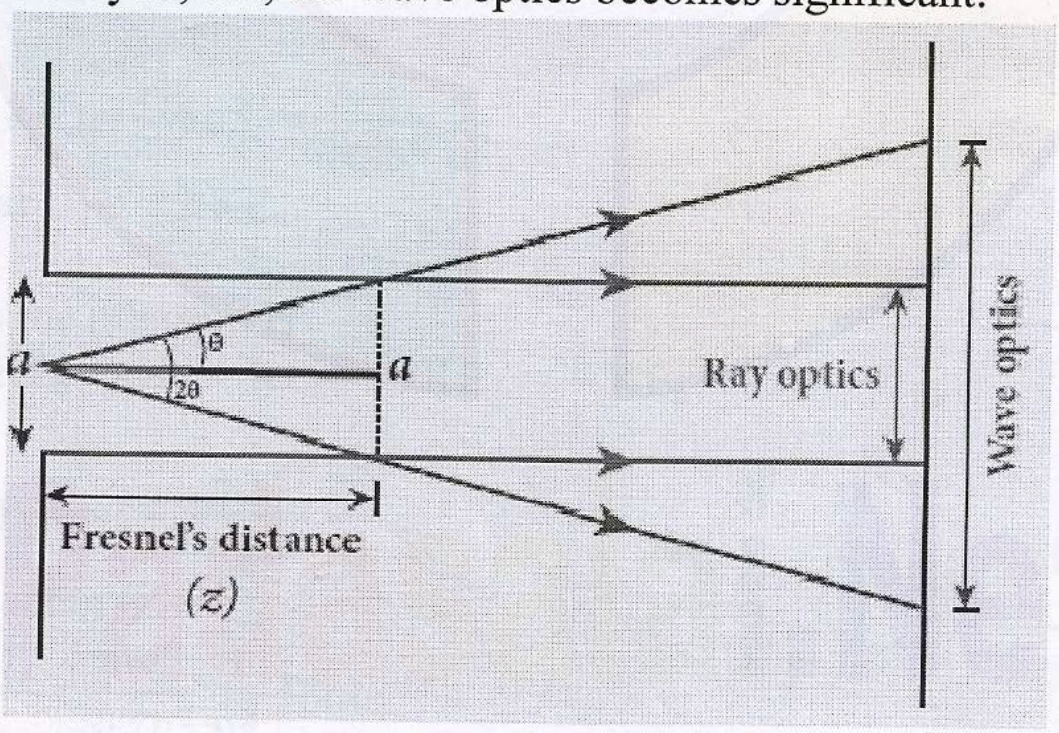
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27. Write a note on energy losses in transformer.

S NO	Name of the loss	Source of loss	Method of minimisation
1	Core loss or Iron loss	i) When transformer core is magnetized and demagnetized some energy is lost in the form of heat.It is known as Hysteresis loss ii) Alternating magnetic flux in the core induces eddy currents in it. The energy loss is due to the flow of eddy current, called eddy current loss	using steel of high **silicon** content in making transformer core using very **thin laminations** of transformer core.
2	Copper loss	When an electric current flows through them, some amount of energy is dissipated due to **Joule heating**. This energy loss is called copper loss	using **wires of larger diameter**
3	Flux loss	Flux leakage happens when the magnetic lines of primary coil are not completely linked with secondary coil. There is some energy loss due to this flux leakage	**winding coils one over the other**

28. Derive an expression for Fresnel distance

Fresnel's distance is the distance upto which the ray optics is obeyed and beyond which the ray optics is not obeyed; but, the wave optics becomes significant.



The diffraction equation for first minimum is, $\sin\theta = \frac{\lambda}{a}$; when θ is small, $\theta = \frac{\lambda}{a}$

From the definition of Fresnel's distance, $2\theta = \frac{a}{z}$ (or) $\theta = \frac{a}{2z}$

Equating the above two equation for θ gives , $\frac{\lambda}{a} = \frac{a}{2z}$

After rearranging, we get Fresnel's distance $z = \frac{a^2}{2\lambda}$

29. A thin converging glass lens made of glass with refractive index 1.5 has a power of + 5.0 D. When this lens is immersed in a liquid of refractive index n, it acts as a divergent lens of focal length 100 cm. What must be the value of n?

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$n_g = 1.5$, power of the converging lens P_1=5.0D, f= −100cm = −m

$n_a = 1$, power of the diverging lens in the liquid $P_2 = \frac{1}{f} = \frac{1}{(-1)} = -1.0$ D

Power of the lens in air $P_1 = (\frac{n_g}{n_a} - 1)(\frac{1}{R_1} - \frac{1}{R_2})$

Power of the lens in liquid $P_2 = (\frac{n_g}{n} - 1)(\frac{1}{R_1} - \frac{1}{R_2})$

$$\frac{P_1}{P_2} = \frac{(\frac{n_g}{n_a} - 1)}{(\frac{n_g}{n} - 1)}$$

$$\frac{5}{-1} = \frac{(\frac{1.5}{1} - 1)}{(\frac{1.5}{n} - 1)} = \frac{(0.5)}{(\frac{1.5}{n} - 1)}$$

$$\frac{1.5}{n} - 1 = \frac{-0.5}{5} = -0.1$$

$$\frac{1.5}{n} = -0.1 + 1 = 0.9$$

$$\frac{1}{n} = \frac{0.9}{1.5} = \frac{3}{5} \qquad \therefore n = \frac{5}{3} = 1.67$$

30. Write construction and working of photo emissive cell.

Construction:

It consists of an evacuated glass or quartz bulb

A cathode and an anode are fixed as shown in Figure

The cathode C is semi-cylindrical in shape and is coated with a photo sensitive material.

The anode A is a thin rod or wire kept along the axis of the semi-cylindrical cathode.

A potential difference is applied between the anode and the cathode through a galvanometer G.

Working:

When cathode is irradiated with suitable radiation, electrons are emitted from it.

These electrons are attracted by anode and hence a current is produced which is measured by the galvanometer.

For a given cathode, the magnitude of the current depends on

i) the intensity to incident radiation and

ii) the potential difference between anode and cathode.

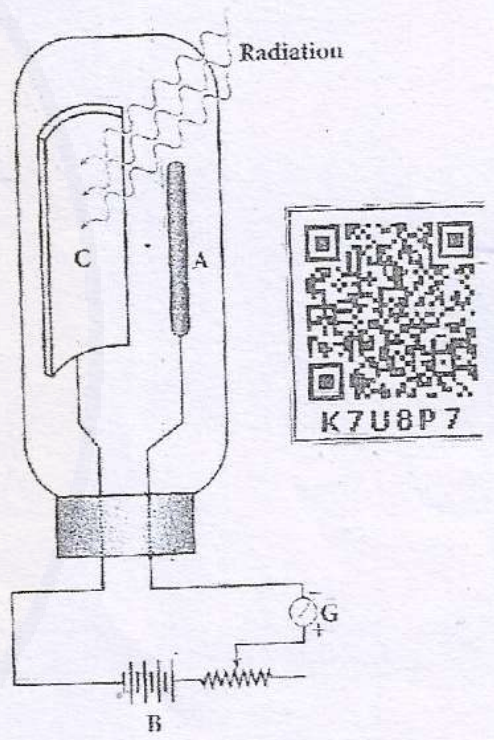


Figure 8.16 Construction of photo cell

31. Show that the mass of radium ($_{88}Ra^{226}$) with an activity of 1 curie is almost a gram.

Given : $T_{1/2}$ = 1600 years.

$\frac{dN}{dt} = 1$ curie $= 3.7 \times 10^{10}$ disintegration / second, $T_{\frac{1}{2}} = 1600$ years

$$\lambda = \frac{0.6931}{1600 \times 365 \times 24 \times 60 \times 60}$$

$$\frac{dN}{dt} = \lambda N \; ; \; N = \frac{dN}{dt} \times \frac{1}{\lambda} = 3.7 \times 10^{10} \times \frac{1600 \times 365 \times 24 \times 60 \times 60}{0.6931} = 2.694 \times 10^{21}$$

6.023×10^{23} atoms in 226 gram

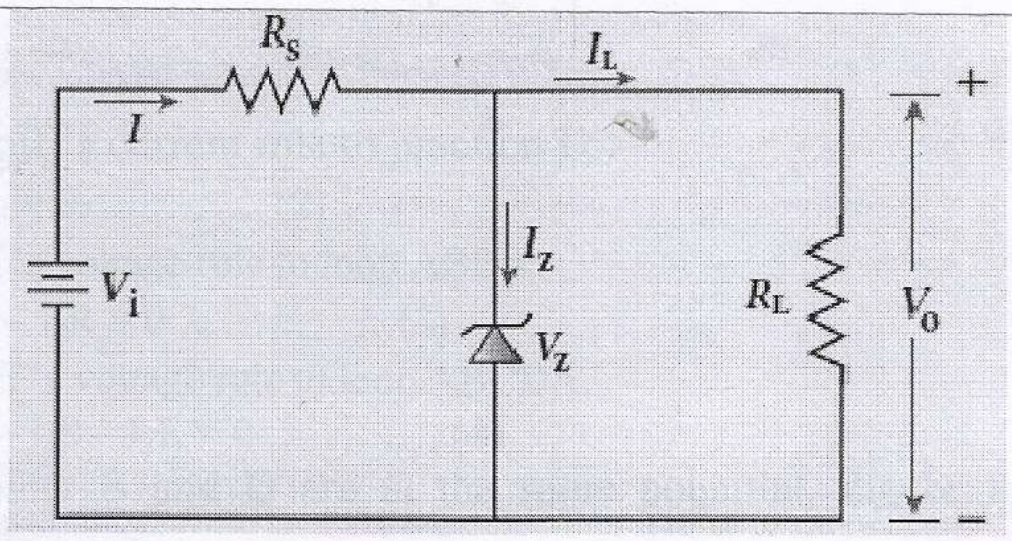
2.694×10^{21} atoms in $= \frac{226 \times 2.694 \times 10^{21}}{6.023 \times 10^{23}} = 1.011\text{g} \approx 1$ gram

32. Explain how a zener diode acts as a voltage regulator.

Zener diode working in the breakdown region can serve as a voltage regulator whose circuit diagram is given in Figure

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Construction:

A series resistance R_s plays a role in voltage regulation.

The load resistance R_L which is connected in parallel with Zener diode

The fluctuating DC input voltage is applied to the circuit and constant output voltage V_o is taken across the load resistance R_L.

The output voltage is maintained constant as long as the input voltage is greater than V_Z.

Working:

Input DC voltage	Zener current I_Z	voltage drop across R_S	Load current I_L	voltage drop across R_L	output voltage V_0
Increases	increase	increase	Constant	Constant	Constant
decreases	decreases	decreases	Constant	Constant	Constant

The load resistance R_L which is connected in parallel with Zener diode, the voltage across R_L is also equal to Zener breakdown voltage which is taken as constant output voltage V_0. If there is any change in input voltage, the voltage drop across R_L changes accordingly. But the voltage across Zener diode or voltage across R_L remains constant. Thus the Zener diode acts as a voltage regulator.

33. What are the advantages and disadvantages of FM.

Advantages of FM

i) In FM, there is a large decrease in noise.

ii) The operating range is quite large.

iii) The transmission efficiency is very high

iv) FM bandwidth covers the entire frequency range.

Due to this, FM radio has better quality compared to AM radio.

Limitations of FM

i) FM requires a much wider channel.

ii) FM transmitters and receivers are more complex and costly.

iii) In FM reception, less area is covered compared to AM.

PART - IV

Answer all the questions $5 \times 5 = 25$

34. a) Describe the function of a transistor as an amplifier with the neat circuit diagram. Sketch the input and output wave form.

Amplification is the process of increasing the signal strength (increase in the amplitude).

A single stage transistor amplifier consists of a NPN transistor, which is connected in the common emitter configuration.

A load resistance R_C is connected in series with the collector circuit to measure the output voltage.

The resistance R_1, R_2 and R_E form the biasing and stabilization circuit.

The capacitor C_1 allows only the AC signal to pass through.

The emitter bypass capacitor C_E provides a low reactance path to the amplified AC signal.

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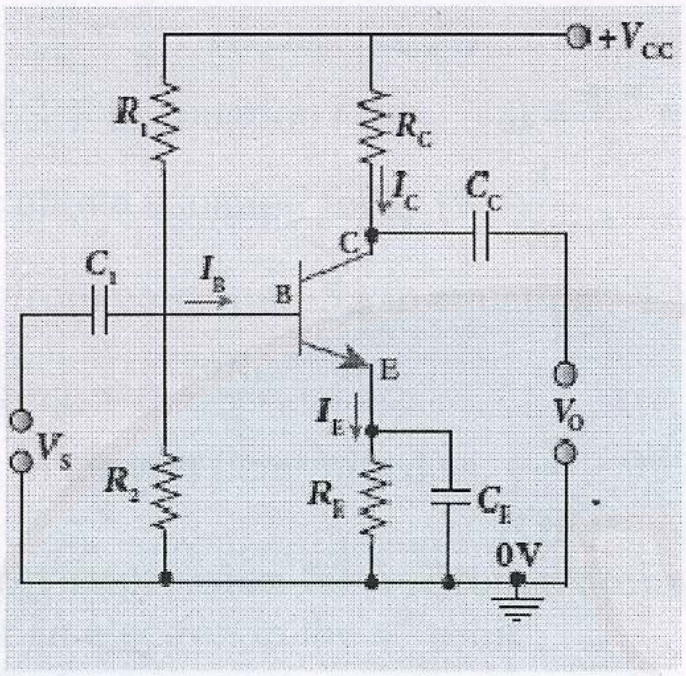
The coupling capacitor C_C is used to couple one stage of the amplifier with the next stage while constructing multistage amplifiers.

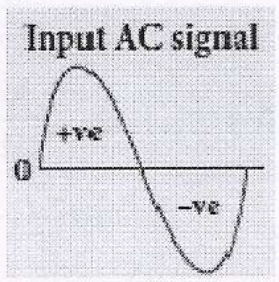
V_S is the sinusoidal input signal source applied across the base-emitter.

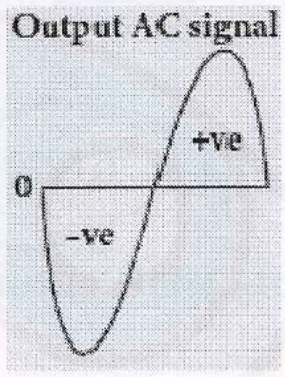
The output is taken across the collector-emitter. Collector current , $I_c = \beta I_B$

Applying Kirchhoff's voltage law to the output loop, the collector-emitter voltage is given by

$$V_{CE} = V_{CC} - I_C R_C$$







Working of the amplifier

During the positive half cycle

Input signal (V_S) increases the forward voltage across the emitter-base. As a result, the base current (I_B in μA) increases. Consequently, the collector current (I_C in mA) increases β times. This increases the voltage drop across R_c (I_cR_c) which in turn decreases the collector-emitter voltage (V_{CE}). Therefore, the input signal during the positive half cycle produces negative half cycle of the amplified signal at the output. Hence, the output signal is reversed by 180°

During the negative half cycle

Input signal (V_S) decreases the forward voltage across the emitter-base. As a result, base current (I_B in μA) decreases and in turn decreases the collector current (I_C in mA). The decrease in collector current (I_C) decreases the potential drop across R_C which in turn increases the collector-emitter voltage (V_{CE}). Thus, the input signal during the negative half cycle produces positive half cycle of the amplified signal at the output.Therefore, 180^0 phase reversal is observed during the negative half cycle of the input signal also as shown in Figure

(OR)

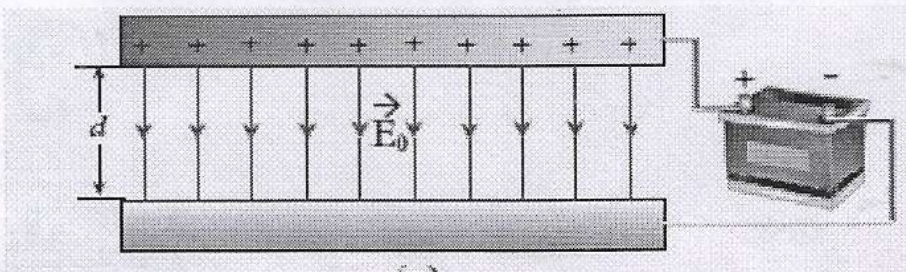
b) Explain the effect of a dielectric placed in a parallel plate Capacitor

Dielectrics like mica, glass or paper are introduced between the parallel plates of a capacitor, then the capacitance of the capacitor is altered.

(i) when the capacitor is disconnected from the battery

Consider a capacitor with two parallel plates each of cross-sectional area A and are separated by a distance d. The capacitor is charged by a battery of voltage V_o and the charge stored is Q_o.

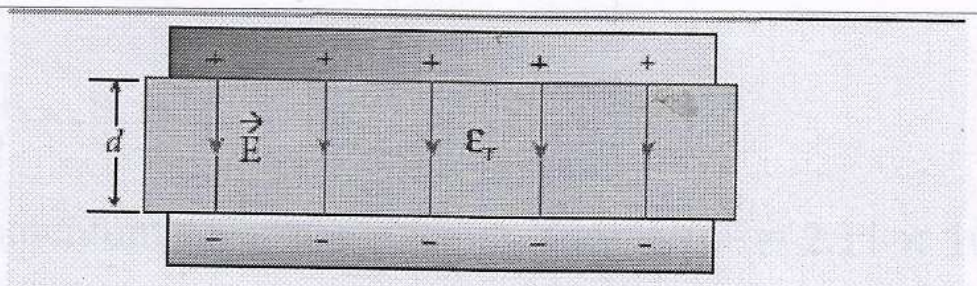
The capacitance of the capacitor without the dielectric is $C_o = \frac{Q_O}{V_O}$



(a)

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(b)

The battery is then disconnected from the capacitor and the dielectric is inserted between the plates.

The introduction of dielectric between the plates will **decrease the electric field**.

Experimentally it is found that the modified electric field is given by

$$E = \frac{E_O}{\epsilon_r}$$

where E_o is the electric field inside the capacitors when there is no dielectric and ϵ_r is the relative permittivity of the dielectric or simply known as the dielectric constant.

Since $\epsilon_r > 1$, the electric field $E < E_o$.

As a result, the electrostatic potential difference between the plates ($V = Ed$) is also reduced. But at the same time, the charge Q_o will remain constant once the battery is disconnected.

Hence the new potential difference is

$$V = Ed = \frac{E_o}{\epsilon_r} d = \frac{V_O}{\epsilon_r}$$

We know that capacitance is inversely proportional to the potential difference. Therefore as **V decreases, C increases.**

Thus new capacitance in the presence of a dielectric is

$$C = \frac{Q_O}{V} = \epsilon_r \frac{Q_O}{V_O} = \epsilon_r C_O$$

Since $\epsilon_r > 1$, we have $C > C_o$. Thus insertion of the dielectric increases the capacitance.

$$C = \frac{\epsilon_r \epsilon_o A}{d} = \frac{\epsilon A}{d}$$

where $\epsilon = \epsilon_r \epsilon_o$ is the permittivity of the dielectric medium.

The energy stored in the capacitor before the insertion of a dielectric is given by

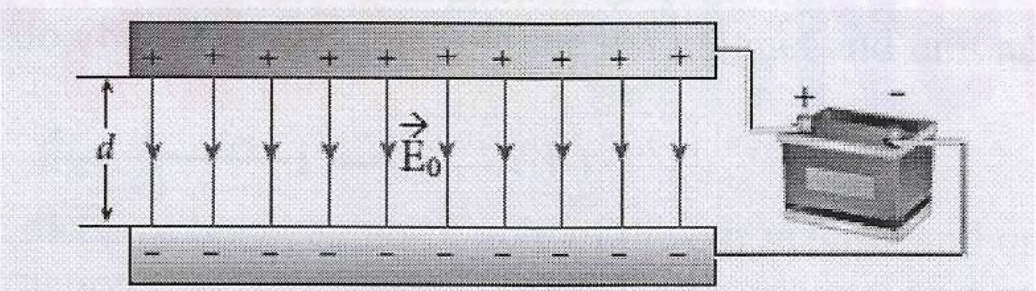
$$U_O = \frac{1}{2} \frac{Q_O^2}{C_O}$$

After the dielectric is inserted, the charge Q_0 remains constant but the **capacitance is increased.** As a result, the stored energy is decreased.

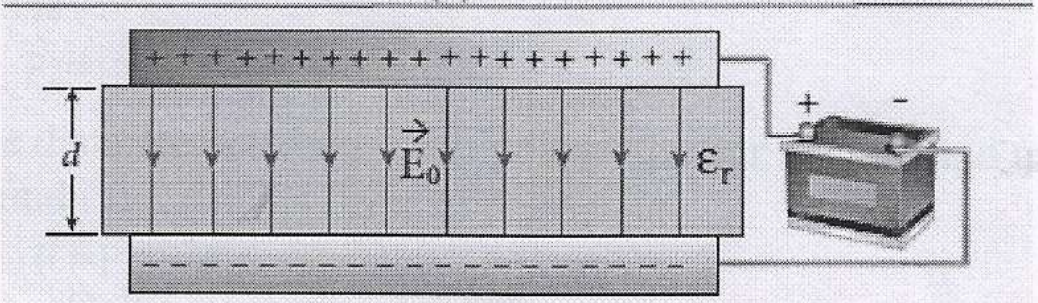
$$U = \frac{1}{2} \frac{Q_O^2}{C} = \frac{1}{2} \frac{Q_O^2}{\epsilon_r C_O} = \frac{U_o}{\epsilon_r}$$

Since $\epsilon_r > 1$ we get $U < U_o$. There is a **decrease in energy**

(ii) When the battery remains connected to the capacitor



(a)



(b)

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Let us now consider what happens when the battery of voltage V_o remains connected to the capacitor when the dielectric is inserted into the capacitor. This is shown in Figure.

The potential difference V_o across the plates remains constant. But it is found experimentally (first shown by Faraday) that when dielectric is inserted, the **charge** stored in the capacitor is **increased** by a factor ϵ_r.

$$Q = \epsilon_r Q_o$$

Due to this increased charge, the capacitance is also increased. The new capacitance is

$$C = \frac{Q}{V_O} = \epsilon_r \frac{Q_O}{V_O} = \epsilon_r C_O$$

However the reason for the increase in capacitance in this case when the battery remains connected is different from the case when the battery is disconnected before introducing the dielectric.

$$\text{Now, } C_o = \frac{\epsilon_o A}{d} \text{ and } C = \frac{\epsilon A}{d}$$

The energy stored in the capacitor before the insertion of a dielectric is given by

$$U_O = \frac{1}{2} C_o V_O^2$$

After the dielectric is inserted, the **capacitance is increased**; hence the **stored energy is also increased.**

$$U = \frac{1}{2} C V_O^2 = \frac{1}{2} \epsilon_r C_o V_O^2 = \epsilon_r U_o$$

Since $\epsilon_r > 1$ we have $U > U_o$.

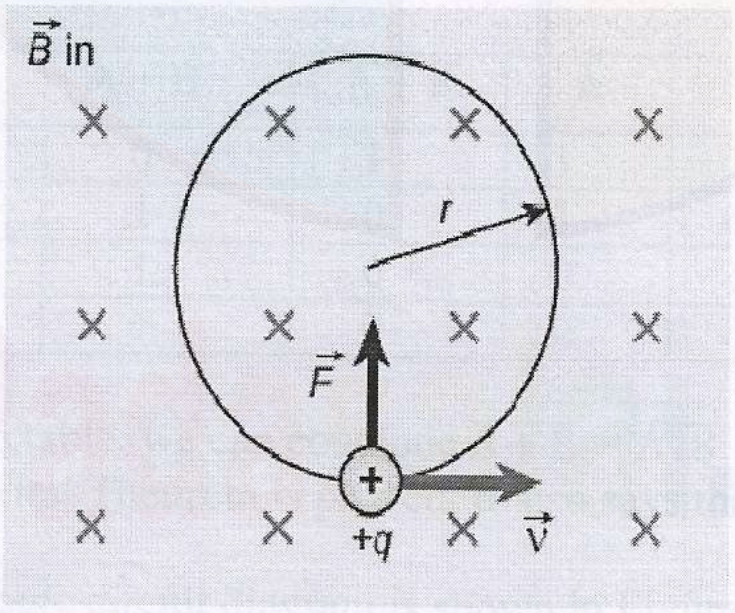
It may be noted here that since voltage between the capacitor **V_0 is constant**, the electric field between the plates also remains constant.

35. a) i) Explain the motion of the charged particle in a uniform magnetic field

Consider a charged particle of charge q and mass m enters into a region of uniform magnetic field $\vec{B}$ with velocity $\vec{v}$ such that velocity is perpendicular to the magnetic field.

When the particle enters into the field, Lorentz force acts on it in a direction perpendicular to both magnetic field $\vec{B}$ and $\vec{v}$ velocity

As a result, the charged particle moves in a circular orbit as shown in Figure



The Lorentz force on the charged particle is given by

$$\vec{F} = q(\vec{v} \times \vec{B}) \text{----------(1)}$$

Since Lorentz force alone acts on the particle, the magnitude of the net force on the particle is

$$\sum F_i = F_m = qvB \text{ ----------(2)}$$

This Lorentz force acts as centripetal force for the particle to execute circular motion.

$$\text{Therefore, } qvB = m\frac{v^2}{r} \text{ ----------(3)}$$

$$\text{The radius of the circular path is } r = \frac{mv}{qB} = \frac{p}{qB} \text{ ----------(4)}$$

where p = mv is the magnitude of the linear momentum of the particle.

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Let T be the time taken by the particle to finish one complete circular motion, then

$$T = \frac{2\pi r}{v} \text{ ----------(5)}$$

Hence substituting (4) in (5), we get

$$T = \frac{2\pi m}{qB} \text{ ----------(6)}$$

Equation (6) is called the **cyclotron period**.

The reciprocal of time period is the frequency *f*, which is

$$f = \frac{1}{T} = \frac{qB}{2\pi m} \text{ ----------(7)}$$

In terms of angular frequency $\omega = 2\pi f = \frac{q}{m} B$ -------(8)

Equations (7) and (8) are called as **cyclotron frequency or gyro-frequency**.

From equations (6), (7) and (8), we infer that time period and frequency depend only on charge-to-mass ratio (specific charge) but not velocity or the radius of the circular path.

ii) An electron moving perpendicular to a uniform magnetic field 0.5 T undergoes circular motion of radius 2.5 mm .What is the speed of electron?

$$Bqv = \frac{mv^2}{r}$$

$$\text{Speed of the electron } v = |q| \frac{Br}{m}$$

$$= 1.6\times 10^{-19} \times \frac{0.5 \times 2.5\times 10^{-3}}{9.11 \times 10^{-31}}$$

$$= \frac{2 \times 10^9}{9.11} = 0.2195 \times 10^9$$

$$= 2.195 \times 10^8 \text{ ms}^{-1}$$

(OR)

b) i) Derive an expression for lateral magnification in spherical mirrors.

Lateral magnification in spherical mirrors

The lateral or transverse magnification is defined as the ratio of the height of the image to the height of the object. The heigh of the object and image are measured perpendicular to the principal axis.

$$\text{Magnification (m)} = \frac{\text{height of the image}(h')}{\text{height of the object}(h)}$$

$$m = \frac{h'}{h}$$

Applying proper sign conventions for equation (6.5),

$$\frac{A'B'}{AB} = \frac{PA'}{PA}$$

$A'B' = -h,\ AB = h,\ PA' = -v, PA = -u$

$$\frac{-h'}{h} = \frac{-v}{-u}$$

simplifying we get,

$$m = \frac{h'}{h} = -\frac{v}{u}$$

Using mirror equation, we can further write the magnification as,

$$m = \frac{h'}{h} = \frac{f-v}{f} = \frac{f}{f-u}$$

ii) An object is placed at a distance of 20.0 cm from a concave mirror of focal length 15.0 cm. What distance from the mirror a screen should be placed to get a sharp image? What is the nature of the image?

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f = – 15cm, u = – 20cm, v = ?

i) From mirror equation $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$; $\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{u-f}{fu}$

$$\therefore v = \frac{uf}{u-f} = \frac{(-20)(-15)}{(-20)-(-15)} = \frac{300}{-20+15} = \frac{300}{-5} = -60 \text{ cm}$$

place the screen at 60 cm , to the left side of the concave lens.

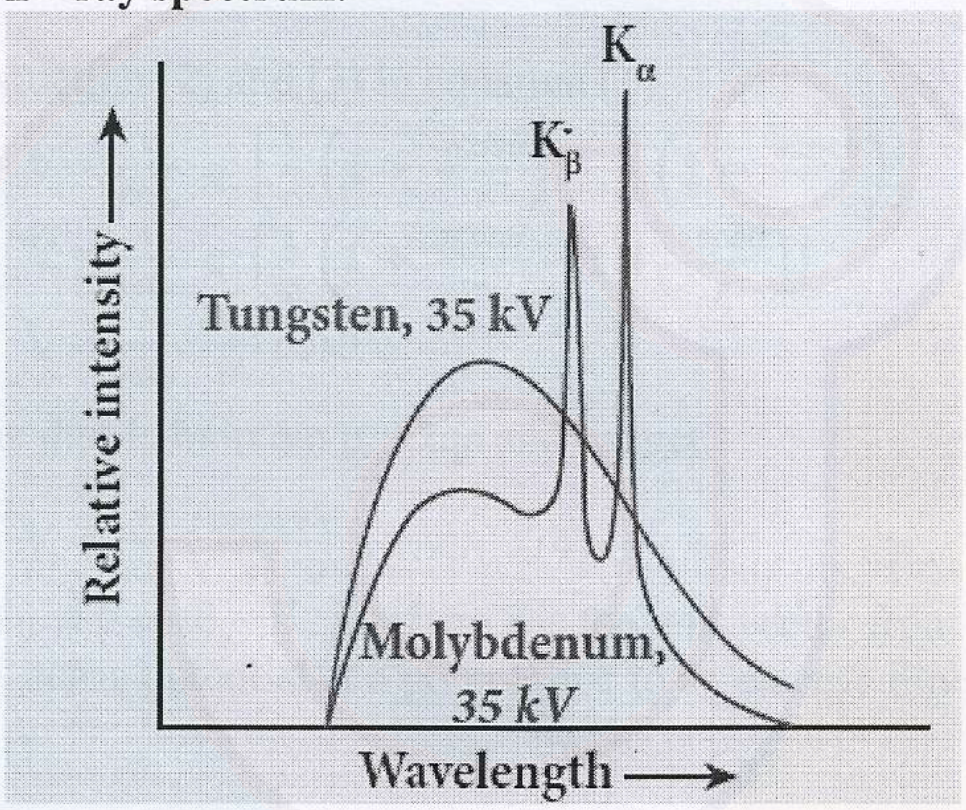
ii) magnification $m = \frac{h'}{h} = -\frac{v}{u} = -\frac{(-60)}{(-20)} = -3$

magnification is negative, hence we get the inverted image

magnification is 3, **image three times greater and real image.**

36. a) Write a note on characteristic X – rays. Give the applications of X – rays in medical therapy and industry.

when the target is hit by fast electrons, X – ray spectra consists of some narrow peaks. The peaks are characteristics of the material of the target and hence it is called **Characteristic x – ray spectrum.**



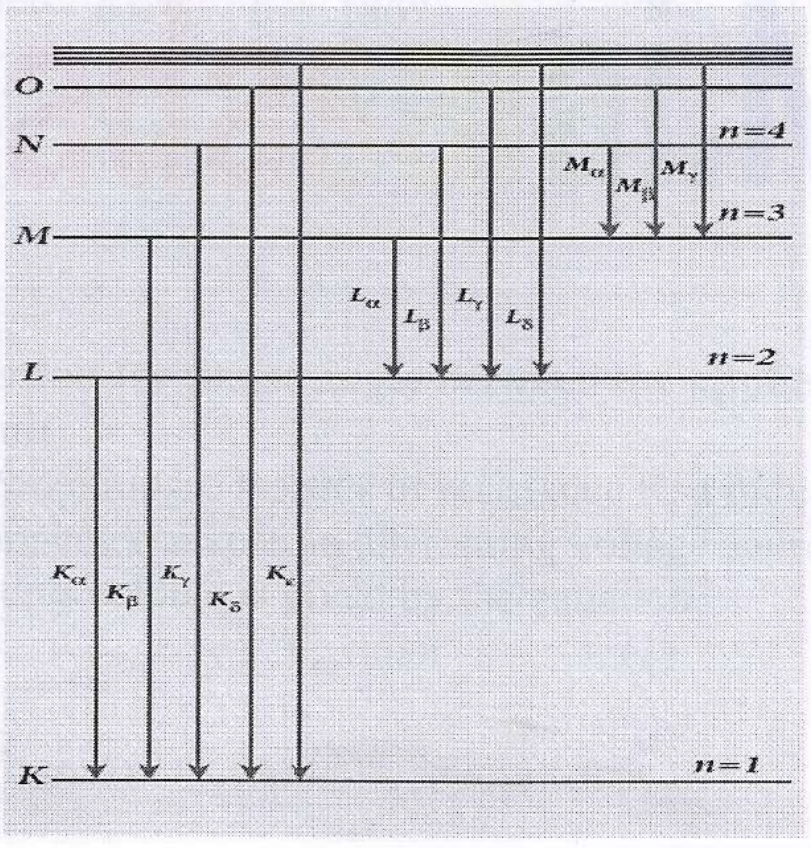
When an energetic electron penetrates into the target atom and it can remove some of the electrons in K-shell and create a vacancy in it.

The energy difference between the levels is given out in the form of x– ray photon of definite wavelength.

Due to the electronic transitions from L, M, N,.. levels to the K-level, K-series of lines $K_\alpha, K_\beta, K_\gamma$….. are obtained.

The K_α and K_β of the K-series of molybdenum are shown in Figure

Due to the electronic transitions from M, N,O . . levels to the L-level, we get L-series and so on



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Applications of X – rays

1) **Medical therapy**

x-rays can kill diseased tissues, they are employed to cure skin diseases, malignant tumours etc.

2) **Industry**

X-rays are used to check for flaws in welded joints, motor tyres, tennis balls and wood. At the custom post, they are used for detection of contraband goods.

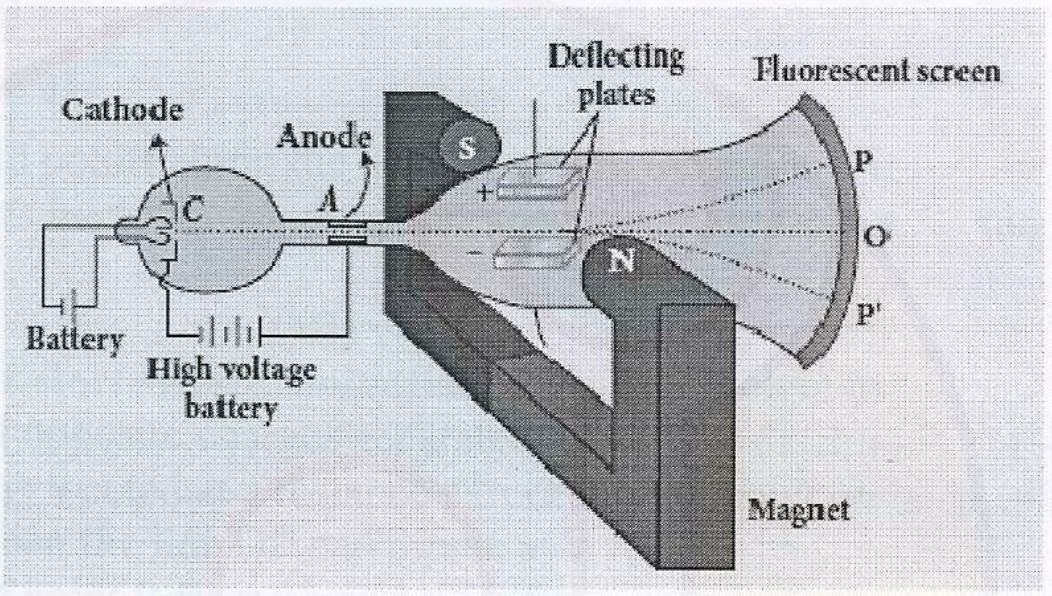
(OR)

b) Explain the J.J. Thomson experiment to determine the specific charge of electron.

Specific charge : charge per unit mass

Principle: the cathode rays were deflected by electric and magnetic fields

Construction:



i) A highly evacuated discharge tube is used

ii) cathode rays produced at cathode are attracted towards anode disc A.

iii) Pin hole in the anode disc is allow only a narrow beam of cathode rays.

iv) Electric field is provided by parallel metal plates with high voltage and the magnetic field is provided by the magnets. Here both electric and magnetic fields are acting perpendicular to each other.

v) When the cathode rays strike the screen, they produce scintillation observed.

vi) This is achieved by coating the screen with zinc sulphide.

(i) Determination of velocity of cathode rays

For a fixed electric field between the plates, the magnetic field is adjusted such that

the cathode rays strike at the original position O

$\therefore$ the magnitude of electric force = the magnitude of force due to magnetic field

$$eE = eBv$$

$$\boxed{v = \frac{E}{B}}$$

(ii) Determination of specific charge

Since the cathode rays are accelerated from cathode to anode,

the potential energy of the electron beam is converted into kinetic energy.

Then from law of conservation of energy,

$$eV = \frac{1}{2} mv^2$$

$$\frac{e}{m} = \frac{v^2}{2V} \text{ ----------(2)}$$

Substituting (1) in (2)

$$\frac{e}{m} = \frac{1}{2V} \frac{E^2}{B^2} \text{ --------(3)}$$

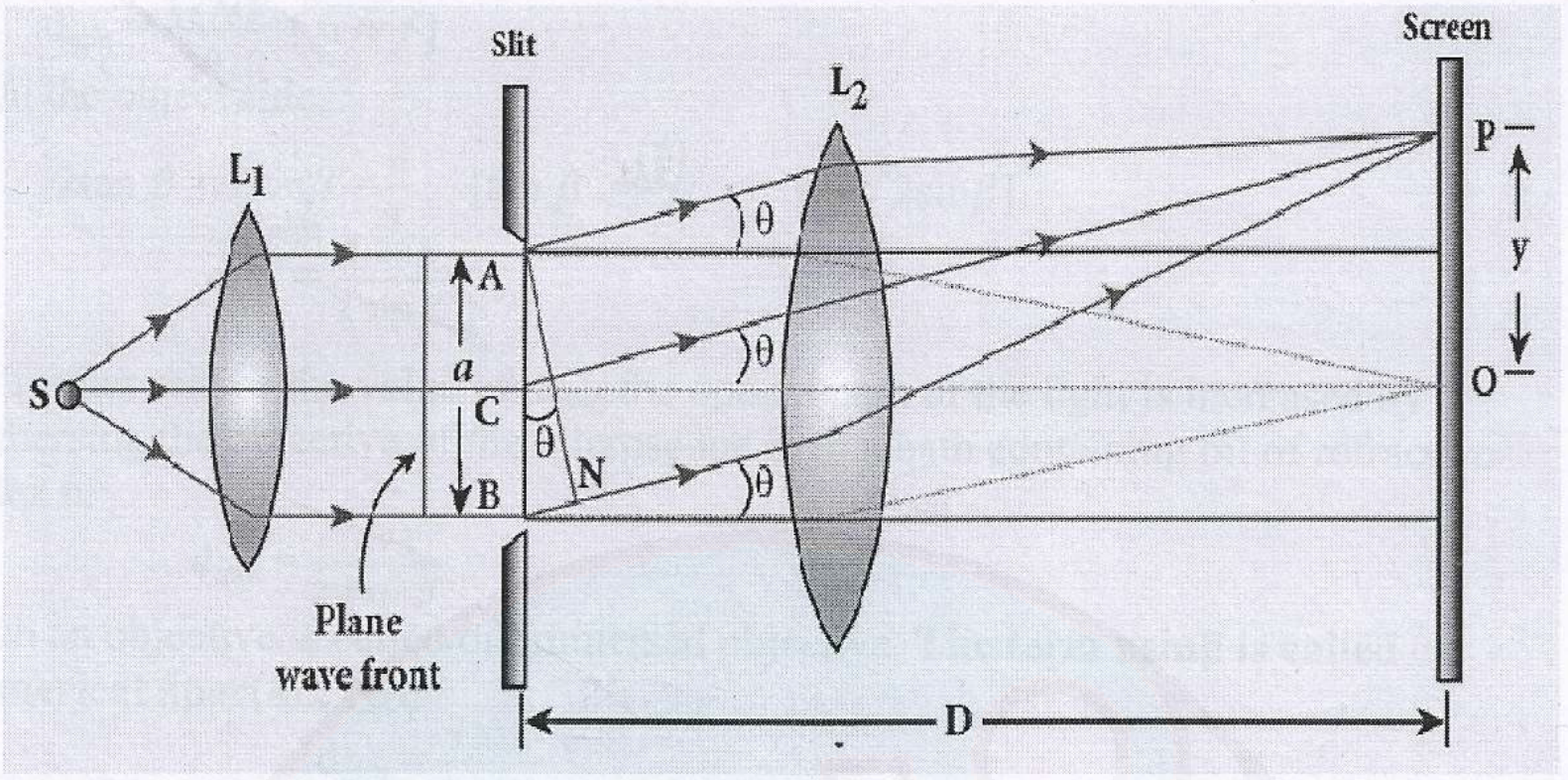
Substituting the values of E, B and V, the specific charge can be determined as

$$\frac{e}{m} = 1.7 \times 10^{11} \text{ C kg}^{-1}$$

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37. a) Discuss diffraction of a single slit and obtain the condition for n^{th} minimum.



Let a parallel beam of light (plane wavefront) fall normally on a single slit AB of width a as shown in Figure

The diffracted beam falls on a screen kept at a distance D from the slit. The center of the slit is C.

All the light reaching the point P make an angle θ with the normal CO.

All the light waves interfere at point P on the screen to give the resultant intensities.

Condition for minimum

Let us divide the slit AB into two halves AC and CB. Now the width of each part is a/2.

The path difference δ between the waves from these corresponding points is $\delta = \frac{a}{2}\sin\theta$

Condition for first minimum at P $\quad a\sin\theta = \lambda$

Condition for second minimum at P $\quad a\sin\theta = 2\lambda$

Condition for n^{th} minimum at P $\quad a\sin\theta = n\lambda$ where n = 1,2,3,……

Condition for maximum

Condition for first maximum at P, $\quad a\sin\theta = \frac{3\lambda}{2}$

Condition for second maximum at P, $\quad a\sin\theta = \frac{5\lambda}{2}$

Condition for n^{th} maximum at P, $\quad a\sin\theta = (2n+1)\frac{\lambda}{2}$ where n = 0,1,2,3,……

(OR)

b) Obtain an expression for average power of AC over a cycle. Discuss its special cases and explain wattles current

Power of a circuit is defined as the rate of consumption of electric energy in that circuit.

It is given by the product of the voltage and current.

In an AC circuit, the voltage and current vary continuously with time.

Let us first calculate the power at an instant and then it is averaged over a complete cycle.

The alternating voltage and alternating current in the series inductive RLC circuit at an instant are given by

$$v = V_m\sin\omega t \ , \ i = I_m\sin(\omega t + \phi)$$

where ϕ is the phase angle between *v and i.*

The instantaneous power is then written as

$$P = v\,i$$
$$= V_m I_m \sin\omega t \sin(\omega t + \phi)$$

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$= V_m I_m \sin\omega t [\sin\omega t \cos\phi + \cos\omega t \sin\phi]$

$= V_m I_m [\sin^2\omega t \cos\phi + \sin\omega t \cos\omega t \sin\phi]$

Here the average of $\sin^2\omega t$ over a cycle is $\frac{1}{2}$ and that of $\sin\omega t \cos\omega t$ is zero.

Substituting these values, we obtain average power over a cycle.

$$P_{av} = V_m I_m \cos\phi \times \frac{1}{2}$$

$$P_{av} = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos\phi$$

$$P_{av} = V_{RMS} I_{RMS} \cos\phi$$

where $V_{RMS} I_{RMS}$ is called apparent power and

$\cos\phi$ is power factor.

The average power of an AC circuit is also known as the true power of the circuit.

Special Cases

(i) For a purely resistive circuit, the phase angle between voltage and current is zero and $\cos\phi = 1$.

$$P_{av} = V_{RMS} I_{RMS}$$

(ii) For a purely inductive or capacitive circuit, the phase angle is $\pm\frac{\pi}{2}$ and $\cos(\pm\frac{\pi}{2}) = 0$

$$P_{av} = 0$$

(iii) For series RLC circuit, the phase angle

$$\Phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$$

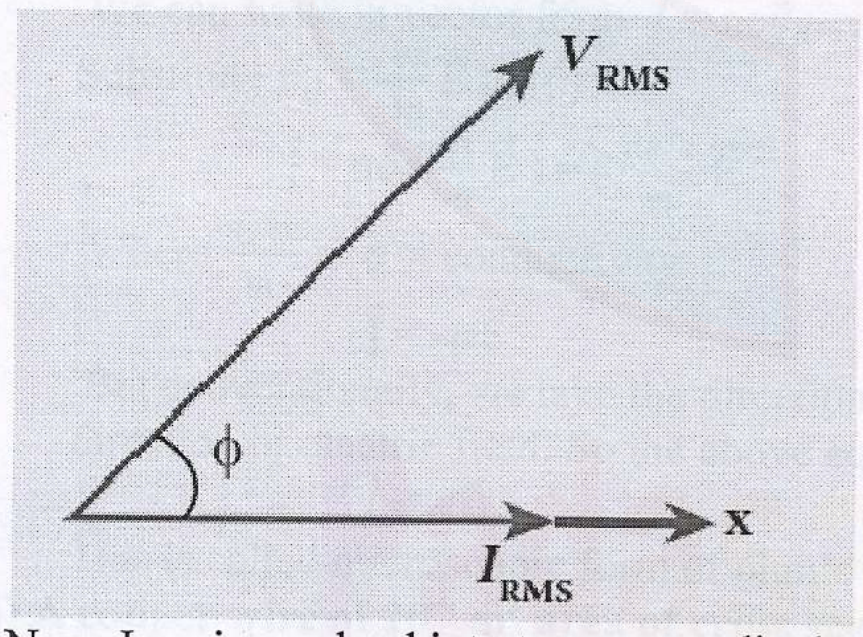
$$P_{av} = V_{RMS} I_{RMS} \cos\phi$$

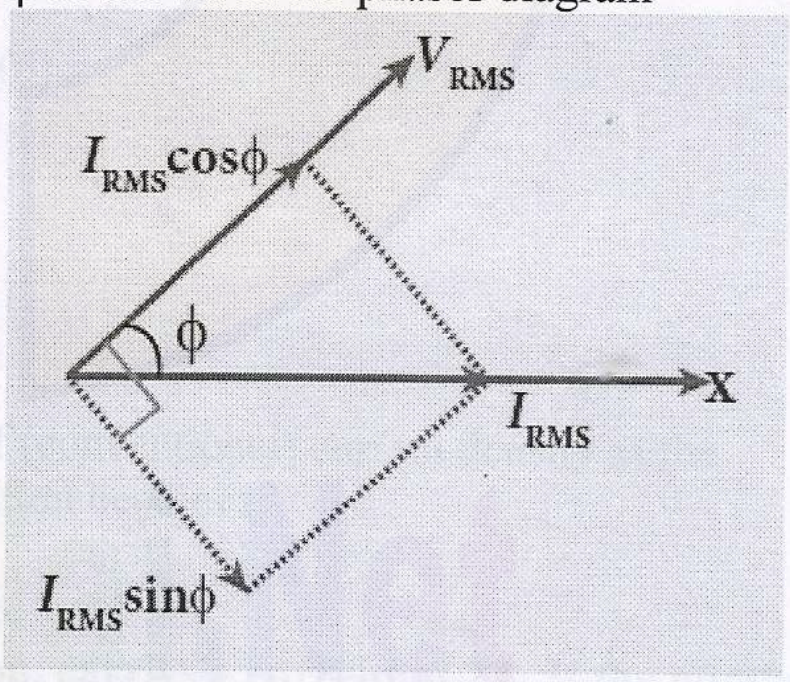
(iv) For series RLC circuit at resonance, the phase angle is zero and $\cos\phi = 1$

$$P_{av} = V_{RMS} I_{RMS}$$

Wattles current:

Consider an AC circuit in which there is a phase angle of ϕ between V_{RMS} and I_{RMS} and voltage is assumed to be leading the current by ϕ as shown in the phasor diagram





Now, I_{RMS} is resolved into two perpendicular components, namely along $I_{RMS}\cos\phi$ along V_{RMS} and $I_{RMS}\sin\phi$ in perpendicular to V_{RMS} as shown in Figure

(i) The component of current ($I_{RMS}\cos\phi$) which is in phase with the voltage is called active component. The power consumed by this current = $V_{RMS} I_{RMS} \cos\phi$. So that it is also known as 'Wattful' current.

(ii) The other component ($I_{RMS}\sin\phi$)which has a phase angle of $\frac{\pi}{2}$ with the voltage is called reactive component. The power consumed is zero. Hence it is also known as 'Wattless' current.

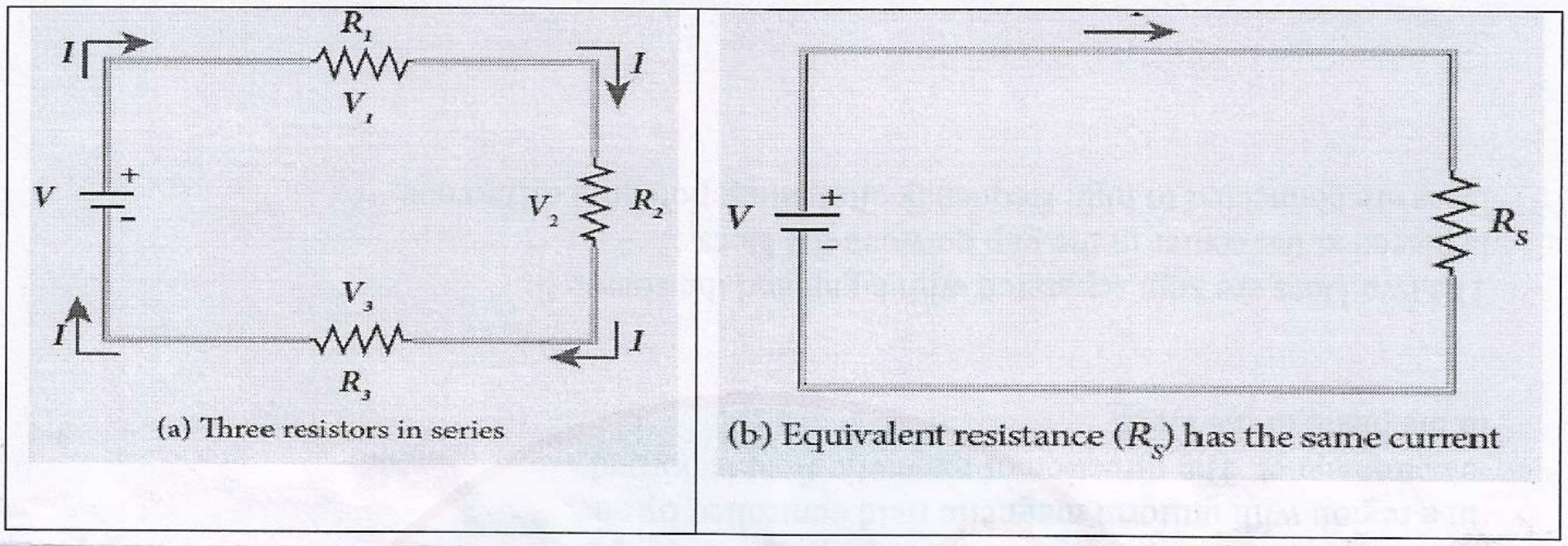
The current in an AC circuit is said to be wattles current if the power consumed by it is zero. This wattless current happens in a purely inductive or capacitive circuit

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38. a) Explain the equivalent resistance of a series and parallel resistor network

Resistance of a series resistor network:



(a) Three resistors in series

(b) Equivalent resistance (R_S) has the same current

When two or more resistors are connected end to end, they are said to be in series.

Three resistors R_1, R_2 and R_3 connected in series.

The amount of charge passing through resistor R_1 must also pass through resistors R_2 and R_3.

Hence the current I passing through all the three resistors is the same.

According to Ohm's law, if the current is same, the potential difference across each resistor must be different.

Let V_1, V_2, V_3 be the potential difference across R_1, R_2, R_3 respectively.

$$V_1 = IR_1 \ , \quad V_2 = IR_2 \ , \quad V_3 = IR_3$$

the total voltage V is equal to the sum of voltages across each resistor.

$$V = V_1 + V_2 + V_3 = IR_1 + IR_2 + IR_3$$

$$V = I\,(R_1 + R_2 + R_3) \ldots\ldots\ldots\ldots (1)$$

$$V = I.R_S \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

where R_S is the equivalent resistance,

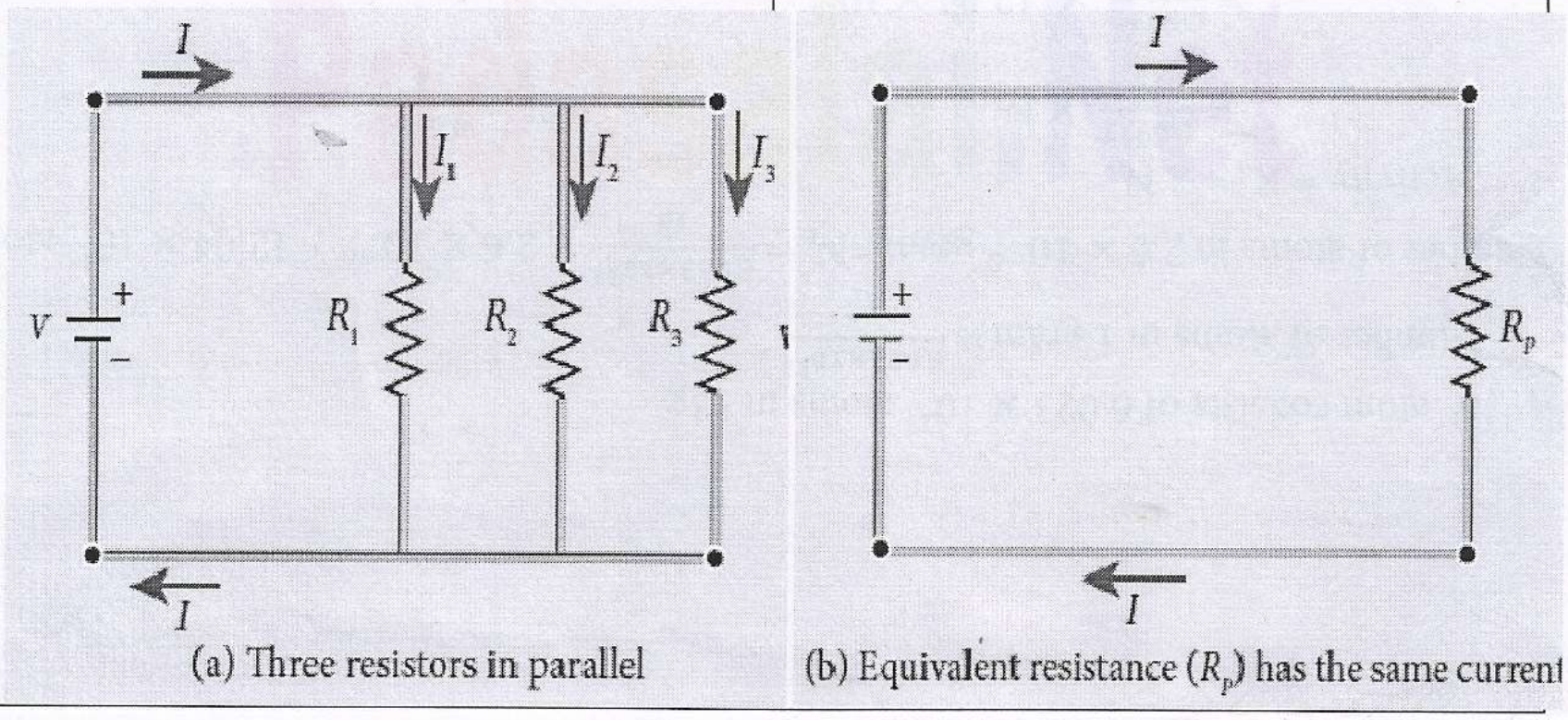
From equation (1) and (2), we have

$$IR_S = I[R_1 + R_2 + R_3]$$

$$R_S = R_1 + R_2 + R_3$$

When several resistances are connected in series, the total or equivalent resistance is the sum of the individual resistances as shown in the Figure (b).

Resistance of a parallel resistor network



(a) Three resistors in parallel

(b) Equivalent resistance (R_p) has the same current

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When two or more resistors are connected across the same potential difference , they are said to be in parallel.

Three resistors R_1, R_2 and R_3 connected in parallel.

In parallel connection the potential difference (V) across each resistor will be the same.

But the current flows through different resistors will be different.

Let I_1, I_2 and I_3 be the current through the resistors R_1, R_2 and R_3 respectively.

Due to the conservation of charge, total current in the circuit I is equal to sum of the currents through each of the three resistors.

$$I = I_1 + I_2 + I_3 \quad \ldots\ldots\ldots\ldots (1)$$

Since the voltage across each resistor is the same, applying Ohm's law to each resistor, we have

$$I_1 = \frac{V}{R_1}; \quad I_2 = \frac{V}{R_2}; \quad I_3 = \frac{V}{R_3}$$

substitute I_1, I_2 and I_3 in (1) $\quad I = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3} =$

$$I = V\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right] \ldots\ldots\ldots\ldots \quad (2)$$

$$I = \frac{V}{R_p} \quad \ldots\ldots\ldots\ldots (3)$$

Here R_p is the equivalent resistance of the parallel combination of the resistors

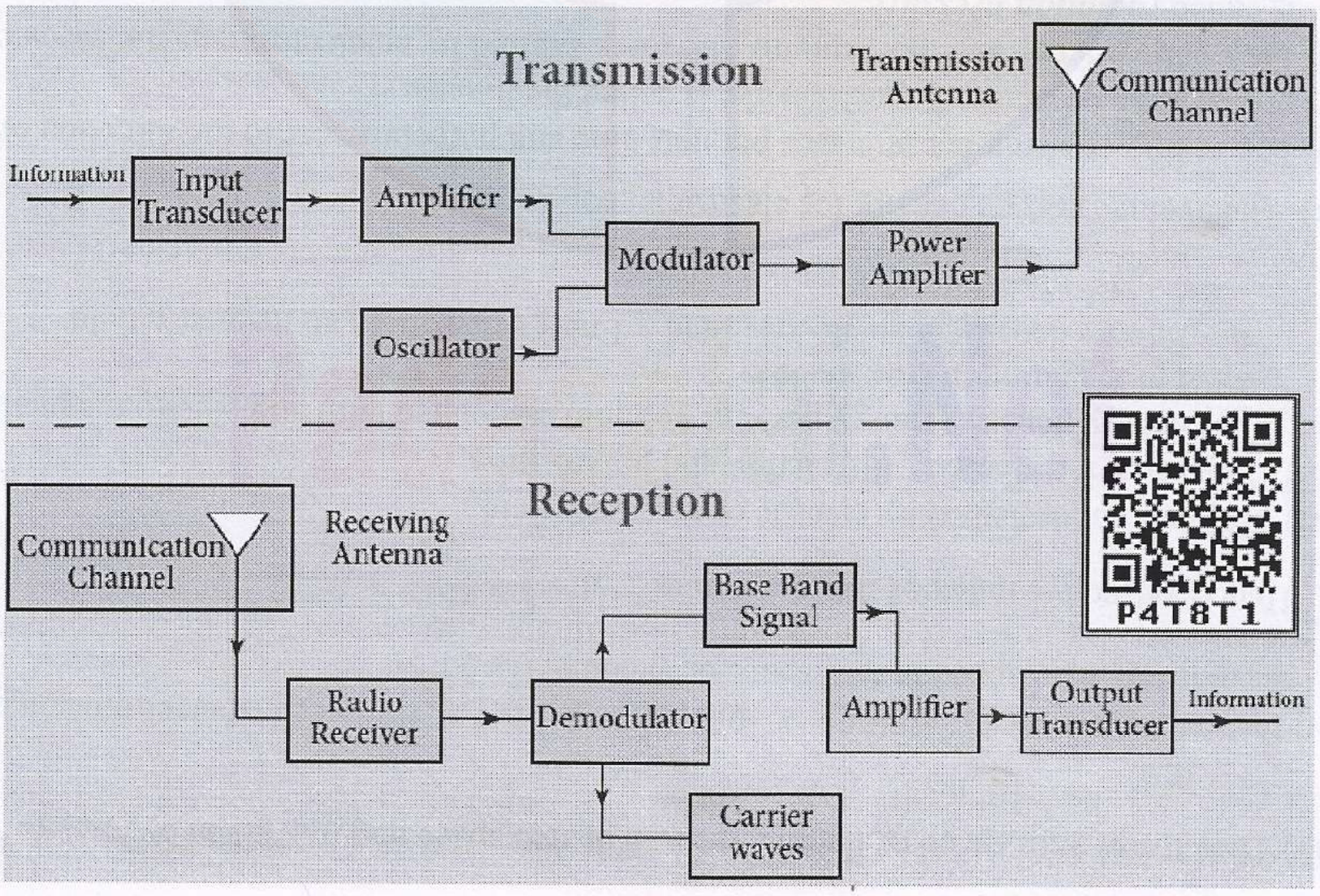
From (2) and (3), $\quad \frac{V}{R_p} = V\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right]$

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

when a number of resistors are connected in parallel, the sum of the reciprocal of the values of resistance of the individual resistor is equal to the reciprocal of the effective resistance of the combination as shown in the Figure (b)

(OR)

b) Explain the elements of electronic communication system.



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i) Information (Baseband or input signal)

Information can be in the form of speech, music, pictures or computer data. This information is given as input to the input transducer.

ii) Input transducer

In communication system, the transducer converts the information into corresponding electrical signals. **The electrical equivalent of the original information is called the baseband signal.**

Example: microphone converts sound energy into electrical energy.

iii) Transmitter

The transmitter is located at the broadcasting station.

Amplifier: The transducer output is very weak and is amplified by the amplifier.

Oscillator: It generates high-frequency carrier wave (a sinusoidal wave) for long distance transmission into space.

Modulator: It superimposes the baseband signal onto the carrier signal and generates the modulated signal.

Power amplifier: It increases the power level of the electrical signal in order to cover a large distance.

iv) Transmitting antenna

It radiates the radio signal into space in all directions. It travels in the form of electromagnetic waves with the speed of light.

v) Communication channel

Communication channel is used to carry the electrical signal from transmitter to receiver with less noise or distortion.

vi) Receiver

The receiving antenna which converts EM waves into RF signals and are fed into the receiver.

The receiver consists of electronic circuits like demodulator, amplifier etc.

The demodulator extracts the baseband signal from the modulated signal. Then the baseband signal is amplified using amplifier. Finally, it is fed to the output transducer.

vii) Repeaters

Repeaters are used to increase the range or distance through which the signals are sent. It is a combination of transmitter and receiver.

The best example is the communication satellite in space.

viii) Output transducer

It converts the electrical signal back to its original form such as sound, music, pictures or data. Examples of output transducers are loudspeakers, picture tubes, computer monitor etc.

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